

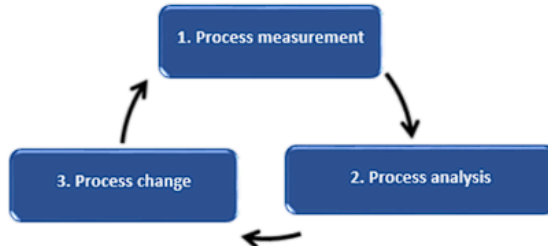
Tutorial 6 - SPM (Wk 11 done)

Tutorial 6: Software Process Improvement

Question 1

Figure 1 illustrates a 3-stage generic framework for improving Software Development Process. Identify THREE (3) types of process metric that you would consider gathering at the beginning of the 3-stage model shown in Figure 1. Provide an example for each of your answers.

Figure 1: Software Process Improvement Cycle



Process Metrics Collection

3 types of process metrics can be collected during process measurement:

1. Time taken for process activities to be completed

✧ *Examples:*

- ✧ *time to complete "requirements gathering",*
- ✧ *time to complete UI design,*
- ✧ *time to complete the software dev. process from start to finish,*
- ✧ *etc*

2. Resources required for processes or activities (e.g. Total effort in **person-days, **hardware** and **software**)**

3. Number of occurrences of a particular event

*e.g.1. no of **defects** discovered 1 week after delivery*

*e.g.2. no of **run-time errors** encountered during 1st UAT*

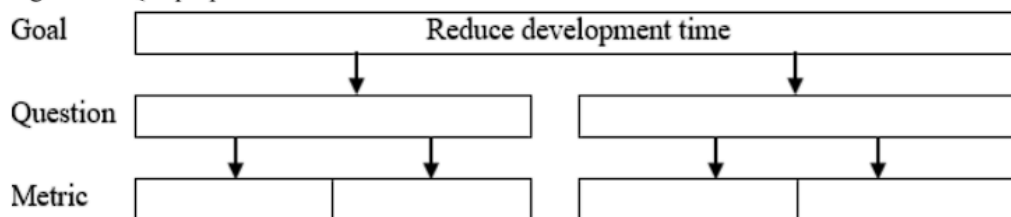
e.g.3. _____

Question 2

You are measuring a software process to solve an issue which is to *reduce development time* from a programmer perspective. It was found that software development time always takes longer than planned. 2 main areas that might contribute to the issue are: time required to collect system requirements with customers and requirement changes. You have identified the metrics in measuring these 2 areas which are time taken for communication with customer, number of communication with customer and pages of requirement documentation. On the other hand, the top management does not agree with your findings and thus will not support your software improvement process (SPI) efforts financially. Seeing that the top management is not supportive, your colleagues appeared less cooperative in giving their feedback when you tried to collect the data for process analysis.

- a) Prepare a Goal-Question-Metric proposal for the given goal by filling the blank in figure 2.
- Analyse TWO (2) questions to assess the given goal.
 - Identify TWO (2) relevant software metrics for each question you have listed. **Explain** each of the software metrics that can be applied to the suggested questions.

Figure 2: GQM proposal



- b) Given below are two difficulties that may arise during process change stage:
- Resistance to change
 - Change Persistence

Select ONE (1) of these difficulties that may arise in the above case scenario. Explain TWO (2) reasons to justify your selections.

a)

Goal: Reduce development time

- Question 1: How much time is required to collect system requirements from customers?**
 - Requirement collection is a critical early stage. If it takes too long, the whole development is delayed. By analyzing the time and effort spent during requirement gathering, we can check if inefficiency here is a root cause of long development time.
- Question 2: How often do requirement changes occur during the project?**
 - Frequent requirement changes cause rework, which increases development time. Monitoring the frequency and scale of these changes helps identify if requirement instability is the main cause of delays.

Metric

For 1st question

- Metric 1: Time taken for communication with customers**

- **Metric 2: Number of communications with customers**

For 2nd question

- Metric 1: Number of requirement change requests
- Metric 2: Number of pages/amount of updated requirement documentation

b) Given below are two difficulties that may arise during process change stage:
- Resistance to change - Change Persistence

Select ONE (1) of these difficulties that may arise in the above case scenario. Explain TWO (2) reasons to justify your selections.

b) Resistance to change

Reason 1: Fear of uncertainty and loss of comfort zone

- Employees and stakeholders are often used to existing workflows, tools, and habits. When a new process is introduced, they may fear that their current skills will become obsolete or that they will fail to adapt to the new expectations. This psychological barrier creates reluctance to accept the change, leading to slowed adoption and even hidden opposition.

Reason 2: Increased workload and training requirements

- Process changes usually require additional training, adjustment to new tools, and a temporary drop in productivity as people learn. Staff may perceive this as extra work added on top of their current responsibilities. Without clear communication of the benefits, they may resist because the short-term burden seems heavier than the long-term gain.

Question 3

Refer to the Figure 1 (Software Process Improvement Cycle) in Question 1, explain TWO (2) techniques used during the process analysis stage.

Extra Ans:

Question 3

Explain TWO (2) techniques used during the process analysis stage.

1. Root Cause Analysis (RCA)
 - Identifies the underlying causes of problems in the process.
 - Example: Using Fishbone Diagram or 5 Whys to analyze why development time is longer than expected.
2. Benchmarking
 - Compares current process metrics against industry standards or best practices.
 - Example: Comparing requirement collection time with average times in similar projects to see if delays are abnormal.

From lecture notes:

a. Questionnaires and interviews

▪ Questionnaires:

✓ **Advantage:** Fast

✓ **Disadvantage:**

- **if questions are inappropriate** □ incomplete/inaccurate understanding of the process
- **Participants** (e.g. engineers and PM) treated it as assessment and **give answer that they think you want to hear.**

▪ Interviews:

✓ **Advantage:** open-ended □ more information

✗ **Disadvantage:** _____

- Answers to a formal **questionnaires** are refined during personal **interviews.**

b. Ethnographic analysis

- Involves assimilating process knowledge by **observation**.
- **Best for in-depth analysis of process fragments rather than for whole-process understanding.**
- The process fragments are chosen from the interview
- Disadvantage:
 - pro-longed process (several months) ☐ from initial stages to maintenance.
 - Impractical for large projects (several years!!??)

